

Leo Murao Watson

Curriculum Vitae

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Education

- Sept 2024 **Statistics Ph.D. Student**, *University of Toronto*.
Uncertainty Quantification and Machine Learning; Supervised by Radu Craiu and Joshua Speagle
- 2020-2024 **BSc., Statistics**, *University of Toronto*, GPA - 3.93.

Selected Graduate-Level Coursework

Machine Learning Methods, Computational Statistics, Stochastic Processes, Quantum Information, Quantum Algorithms, Mathematical Statistics

Reading and Research Projects

- May 2024 - **Decentralized Finance**, *Visiting Researcher, École Normale Supérieure*, Supervisor:
July 2024 *Prof Vincent Danos*.
Worked on optimal allocation problems for automated market makers (paper below). Specifically, developing an algorithm that efficiently computes best execution of a Uniswap V3 market order against a set of liquidity providers expressing their offers as curves.
- Sep 2023 - **Alternatives to Hamiltonian Monte Carlo**, *Reading Course, University of Toronto*,
Dec 2023 *Supervisor: Prof Scott Schwartz*.
Reading course examining properties of modern MCMC methods (Langevin MC, Stochastic Gradient MCMC, etc.) through reading recent papers such as Neal 2012, Betancourt 2017, Nemeth 2019. Writing out step-by-step derivations of claims in papers and repurposing information for use in Prof Schwartz's STA365 Bayesian Statistics course in the form of simulations, write-ups, and interactive notebooks.
- May 2022 - **Predicting Japanese Economic Intervention by Applying NLP to Financial**
Oct 2022 **Texts**, *Research Project, University of Toronto*, Supervisor: *Prof Mark Manger*.
Worked on measuring the semantic polarity of Japanese newspaper articles to predict whether the Japanese government would intervene in the economy. Adapted state-of-the-art Japanese tokenizer library to finance-specific corpora by designing a custom dictionary of terms. Tested resulting tokenizer with multiple word embedding techniques to predict the polarity of articles and aggregated trends over time. Acknowledged as contributor in final published paper ([link](#)).
- May 2022 - **Methods for Missing Data**, *Reading Course, University of Toronto*, Supervisor:
Sep 2022 *Prof Nathalie Moon*.
Reading course following Buuren's *Flexible Imputation of Missing Data*. Created write-up and designed simulations demonstrating the optimality of different imputation techniques under different missingness mechanisms. Presented material weekly and discussed new concepts.

Internship Experience

- June 2023 - **Sales and Trading Internship**, *Citigroup, Tokyo*.
Aug 2023 10 week program rotating at seven desks on the trading floor including:
- **Multi-Asset Structuring**: coded and backtested multi-asset systematic trading strategy based on modern portfolio theory and equity research analyst calls. Ultimately outperformed TOPIX index returns by 75%.
 - **Equity Trading**: Worked on generating alpha by coding cross-market ADR model based on correlations observed in time series data and cluster analysis of similarly behaving equities.
 - **FX Quant**: Analyzed client flow across hundreds of products using matrix decomposition methods to determine client archetypes and estimated flow proportions. Presented to sales team to provide client-tailored trade ideas based on analysis.
- July 2021 - **Quantum Communications Internship**, *Keio University*, *Supervisor: Prof Rodney Van Meter*.
Sep 2021 Through the Q-LEAP program ([link](#)) funded by Japanese Ministry of Education. Developed 180+ pages of lecture notes in English and Japanese for undergraduate course on quantum communications. Wrote Python programs to streamline lecture note creation for future interns. Acknowledged as contributor for the final published textbook: *Quantum Communications* ([link](#)).

Selected Scholarships and Awards

- 2025 NSERC Canada Graduate Scholarship - Doctoral (CGS-D) - \$120,000 over 3 years
2024 Doctoral Recruitment Award - \$10000
2024 Samuel Beatty Scholarship - \$1000
2022 University of Toronto Excellence Award \$7500

Publications and Preprints

- Nov 2024 **Danos, V., Watson, L.M., El Khalloufi, H., Valencia, S.** *On-chain optimal aggregation of Uniswap v3 clones*. Presented at 6th International Conference on Blockchain Economics, Security and Protocols (Tokenomics 2024) ([link](#))

Teaching

- 2025 **UTSSRP Project Mentor**, *University of Toronto*.
Mentored group of four undergraduate students on an applied statistics project over two weeks, culminating in a poster. Topics included causal inference and Bayesian inference.
- 2024–2026 **Teaching Assistant**, *University of Toronto*.
STA130: An Introduction to Statistical Reasoning and Data Science (2 terms)
STA365: Applied Bayesian Statistics
STA255: Statistical Theory

Administrative Service

- Sep 2025 **Local Organizing Committee**, *Fast and Curious 2: MCMC in Action*, *University of Toronto*.
- Sep 2023 - **President**, *Undergraduate Statistics Student Union*, *University of Toronto*.
June 2024 Elected president of 6000+ students in undergraduate statistics programs at the University of Toronto.